RAHUL WALE

AI Engineer · Voice AI & Forward Deployed Solutions

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SUMMARY

AI Engineer building and shipping production LLM and real-time voice AI systems. Scaled conversational platforms to 500+ concurrent users at 99% uptime, cutting operational workload by 60%. Deep hands-on experience with speech-to-speech models (Gemini Live API), real-time voice pipelines (Pipecat, WebSocket audio, VAD, barge-in), RAG systems, and end-to-end ownership from FastAPI backends to React/Next.js frontends.

EXPERIENCE

AI Engineer · The BAAP Company *Jul 2023 – Present*

- Architected and shipped 4+ production AI platforms (real-time voice agents, RAG chatbots, no-code AI tools) serving rural India at 99% uptime under 500+ concurrent users.
- Designed a real-time multilingual Voice-on-Call AI platform combining Gemini LLM, Sarvam STT, Deepgram TTS, and Pipecat over Twilio — cut manual call-centre workload by 60%.
- Scaled a speech-to-speech conversational platform on the Gemini Live API, streaming full-duplex audio over WebSockets directly to a telephony provider — eliminating the separate STT/TTS stack and reducing pipeline latency.
- Built a no-code AI chatbot platform on LangChain + Gemini with RAG over municipal knowledge bases, enabling local governments to self-serve 5,000+ daily citizen interactions with zero engineering support.
- Engineered low-latency voice pipelines (sub-second response) with WebSocket audio streaming, VAD, and barge-in — productionised STT, LLM reasoning, and TTS as a single conversational stack.

KEY PROJECTS

Voice-on-Call AI Platform *Real-time multilingual voice agent for civic services*

Pipecat · Gemini · Sarvam AI · Deepgram · Twilio · FastAPI · React · WebSockets

- Real-time AI voice assistant letting citizens lodge civic queries over a phone call in their native Indian language.
- Owned the full stack: FastAPI backend, React dashboard, and Twilio voice integration; 99% uptime under 500+ concurrent calls.
- Extended the agent with tool calling to take real-world actions — lodging and tracking complaints via structured tool calls with validation, retries, and human handoff on failure.

Multi-Agent Research Assistant *Autonomous text-based agentic system*

LangGraph · Gemini · Tool Calling · Web Search APIs · FastAPI · PostgreSQL

- Multi-agent system where a planner agent decomposes a research query into tasks, parallel worker agents gather and verify sources via tool calls, and a writer agent produces a structured, cited report.
- Implemented agentic orchestration patterns — state management across agents, tool-call validation, retries, and self-correction loops when a worker returns low-quality results.

TECHNICAL SKILLS

- AI / ML: LLMs, Speech-to-Speech Models (Gemini Live), LangChain, LangGraph, RAG, Vector Databases, Embeddings, Gemini, Hugging Face, Prompt Engineering, AI Agents, Tool Calling, LLM Evaluation, Guardrails
- Voice & Real-Time: Pipecat, LiveKit, Sarvam AI, Deepgram, Vosk, VITS, Twilio Voice, WebSocket audio streaming, VAD, barge-in, real-time voice pipelines
- Backend: FastAPI, Python, REST APIs, WebSockets, PostgreSQL, Redis
- Frontend & Mobile: Next.js, React, Flutter (cross-platform iOS / Android)
- Infrastructure: Docker, AWS (EC2, S3), Firebase, Git, CI/CD

EDUCATION

Bachelor of Computer Applications (BCA) *completed 2026*